



Working with Labeled Volumes

A labeling session generates a `groundTruthMedical` Object and labeled volume data. These can be viewed and analyzed outside the app.

Navigate to the Module 3 folder in the course files and work through the **AnalyzingLabeledVolumes.mlx** script. You'll work with a completed labeled volume of the MRI data you saw in the previous video to compute properties like volume for each label.

We encourage you to work through the code in MATLAB. A static file is provided below. The code may be different than in MATLAB because interactive tools are not present in static files.

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Once you have completed these step, uncomment and run the code in the following section.

Load the `groundTruthMed.mat` file from your labeling session folder.

```
% sessionFolder = "";  
% load(fullfile(sessionFolder,"groundTruthMed.mat"))
```

This will load a ground truth medical object named `gtruthMed` into the Workspace. Let's look at the information it contains.

`dataSource`

This field will contain a `Source` for each volume imported into the session. The code below displays the full path of the images in the first volume.

```
% gtruthMed.dataSource.Source{1}
```

`LabelData`

When labels are added to a volume in the app, a new labeled volume is created. The code below displays the full path to this labeled volume.

```
% gtruthMed.LabelData(1)
```

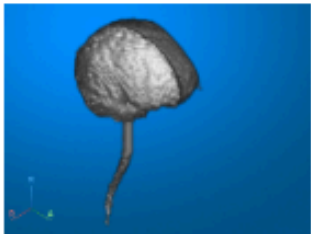
`LabelDefinitions`

Labels definitions must be created in the Medical Image Labeler App in order to label volumes. The `LabelDefinitions` table maps each label name to its corresponding ID.

```
% gtruthMed.LabelDefinitions
```

Inspect Labeled Volume

We used the Medical Image Labeler App to label the left and right hemispheres in the T1-weighted 3 Tesla MRI data. Let's inspect the results using the labeled volume included in the course files.



It can be loaded using `medicalVolume` just like 3D medical images.

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